Kingdom of Saudi Arabia Royal Commission at Yanbu Colleges & Institutes Division		المـمـــــلكـــة العربية الســـعودية بـينبع الـــهـــيـــئة الملـــكـــية قـــطـــاع الكـــــلـيات والـمـعـاهـد
Yanbu Industrial College Computer Science & Engineering Dept. Information & Computer Technology Dept.		كلية ينبع الصناعية قسم علوم وهندسة الحاسب الآلي قسم تقنية المعلومات والحاسب الآلي

PROJECT

ACADEMIC YEAR 1446/1447 H (2025/2026 G), SEMESTER I - 471

DATABASE SYSTEMS
CS 311

STUDENT NAME:	**Wejdan Ahmad - 4211529** **Shahad Ahmed-4311075** **Laila Turki- 4211361** **Juri Omar- 4220348**
STUDENT ID:	**SECTION: 3**
DATE:	**September 30, 2025**

FOR INSTRUCTOR USE ONLY	GENERAL INSTRUCTIONS

Q. No.	CLOs	MAX MARK	MARKS OBTAINED
CD1	1.03 2.02	5	
CD2	2.01 3.01	10	
LD1	2.02 3.01	10	
LD2	3.3	10	
F	3.02	5	
TOTAL MARKS		40	

- **Write your name and ID number in the above space.**
- **Submission is required in soft-copy form (a PDF format - copy-paste codes into this form).**
- **Copied work will not be graded.**

MARKED BY:	**Signature:**
CHECKED BY:	**Signature:**

Project description

The given project is to design a database that keeps all the appropriate information related to any real-time management system. The project will implement both conceptual and logical phases.

The project is divided into two phases.

- Conceptual database design.
- Logical database design.

1. Conceptual Design

1.1. Phase I :

The first part of Conceptual database design is supposed to be discussed in **week 7/8** (as a demonstration (demonstration 1)). Each group of students will discuss the progress, having the following contents.

- A brief introduction about the organization.
- Identification of Business Rules.
- First draft of ERD (entities, relationships)

1.2. Phase II :

The second part of Conceptual database design is supposed to be discussed in **week 10** (as a demonstration (demonstration 2)). Each group of students will discuss the progress, including the following:

- Modifications after instructors' feedback.
- A second draft of ERD (entities with their attributes, relationships with connectivities, and cardinalities).

2. Logical Design

2.1. Phase I :

The first part of logical database design is supposed to be discussed in **week 12** (as a demonstration (demonstration 3)). Each group of students will discuss the progress, having the following contents:

- The detailed process of applying normalization to each entity.
- A final draft of ERD after normalization.

2.2. Phase II :

The second part of the logical database will contain the following contents:

- DDL implementation
- DML implementation
- Query samples, identifying from the database. The queries will be SELECT statements based on multiple conditions, such as AND, OR operators, Joins, and Aggregate functions. The following details are required to submit queries.
    - Statement of queries.
    - Solution of queries written in SQL.
    - Output of queries.

Final Presentation and Report Submission

In **week 14**, each group of students will present their project during class time and submit a written report. The final report will describe all the above-mentioned phases based on the feedback received during the presentation.



Movie Rental Store Database

1.1 Introduction

In this project, we are designing a database system for a Movie Rental Store. The store allows customers to browse available movies, rent them for a limited period, and return them before the due date to avoid late fees. The system aims to organize customer information, rental transactions, movie details, and business rules that control the rental process.

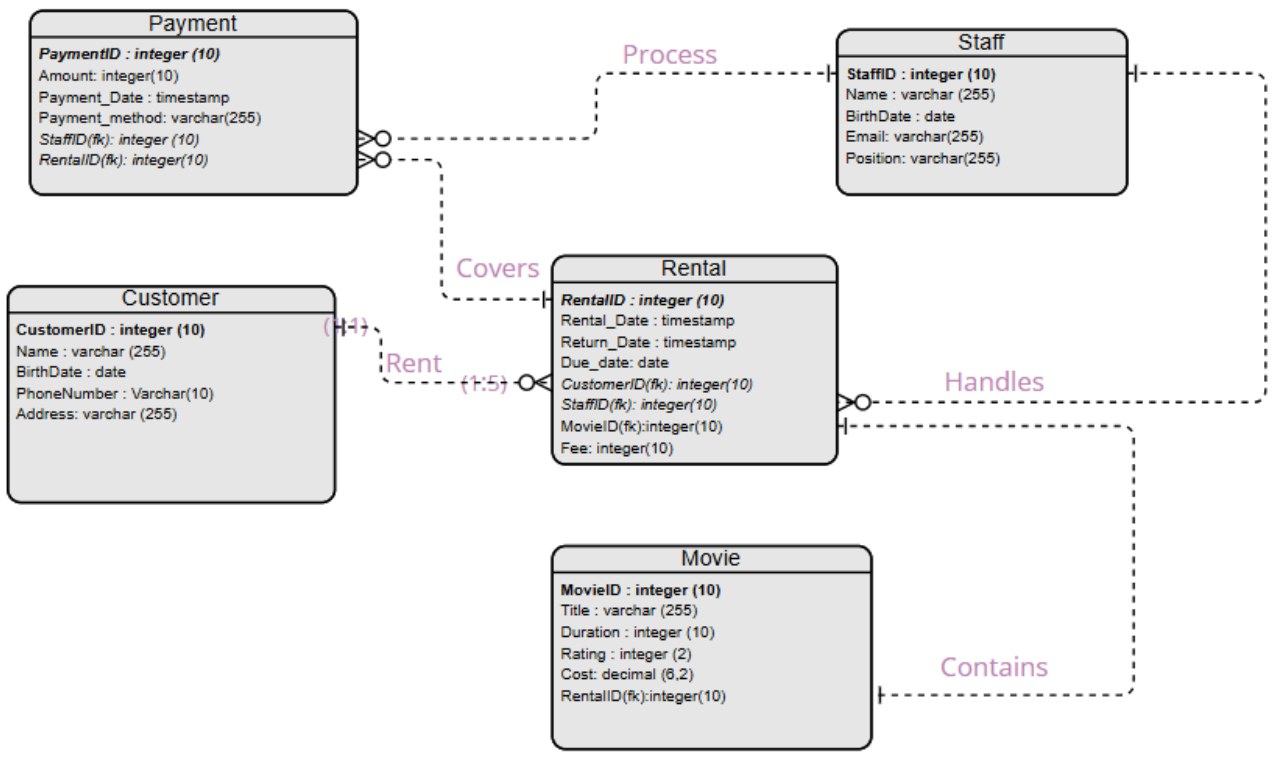
1.2 Objective of The System

The objective of this project is to design a conceptual database model for a Movie Rental Store that supports efficient management of customers, movies, and rental transactions. The system aims to ensure accurate tracking of active rentals, automate late fee calculations, and enforce business rules to maintain smooth and consistent operations.

1.3 Business Rules

1. Customers must be 18 years old or older in order to create an account and rent movies.
2. A customer can have up to five active rentals at the same time.
3. Each movie can be rented for a maximum of 10 days.
4. A late fee of 5 SAR per day will be applied if the customer returns the movie after the due date.

1.4 Initial Entity Relationship Diagram



Entities and attributes:

Customer:

CustomerID (PK)	Name	BirthDate	PhoneNumber	Address

Movie:

MovieID (PK)	Title	Duration	Rating	Cost

Staff:

StaffID (PK)	Name	BirthDate	Email	Position

Rental:

RentalID (PK):	Rental_Date	Return_Date	Due_Date	CustomerID (FK)	StaffID (FK)	Fee

Rental Details :

Rental_detailID (PK)	MovieID (FK)	RentalID (FK)	Description

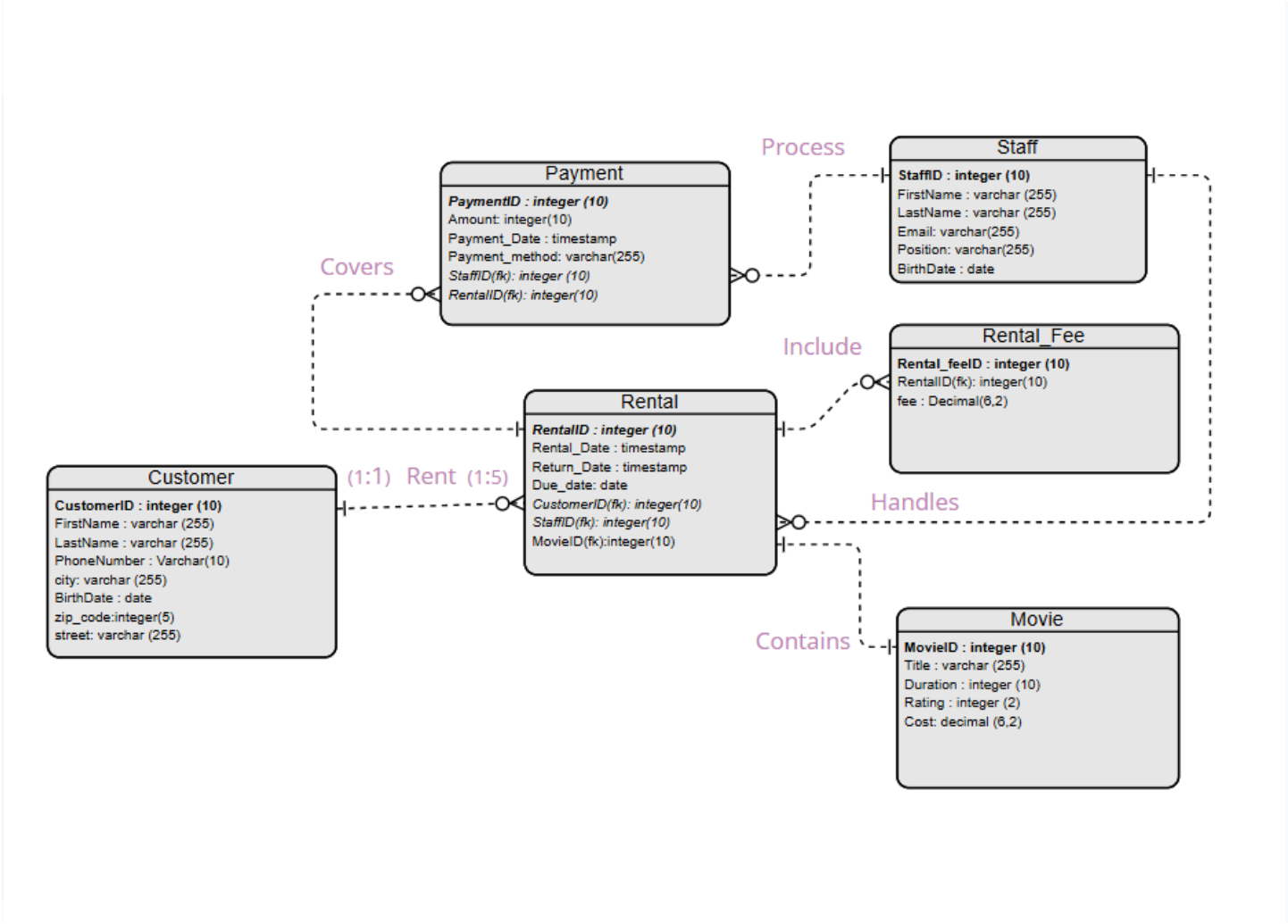
Payment:

PaymentID (PK)	Amount	Payment_Date	Payment_method	StaffID (FK)	RentalID (FK)

Relationships:

- **Customer — Rental** (Rent) Type: One-to-Many (1:M)
- **Staff — Rental** (Covers) Type: One-to-Many (1:M)
- **Staff — Payment** (Process) Type: One-to-Many (1:M)
- **Rental — Payment** (Includes) Type: One-to-Many (1:M)
- **Movie — Rental_Details** (Contains) Type: One-to-Many (1:M)
- **Rental — Rental_Details** (Handles) Type: One-to-Many (1:M)

2.1 Final Entity Relationship Diagram (after normalization):



Normalization

1. **Staff Entity**

Staff (Before Normalization)				
StaffID	Name	BirthDate	Email	Position

Staff (After Normalization)					
StaffID (PK)	FirstName	LastName	BirthDate	Email	Position

1NF: PK/ StaffID

Name contains FirstName, LastName

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, No transitive dependencies, it is in 3NF.

2. **Customer Entity**

Customer (Before Normalization)				
CustomerID	Name	BirthDate	PhoneNumber	Address

Customer (After Normalization)							
CustomerID (PK)	FirstName	LastName	BirthDate	PhoneNumber	Street	City	Zip_code

1NF: PK/ CustomerID

Name contains FirstName, LastName

Address contains Street, City, Zip_code

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, No transitive dependencies, it is in 3NF.

3. Movie Entity

Movie (Before Normalization)				
MovieID	Title	Duration	Rating	Cost

Movie (After Normalization)				
MovieID(PK)	Title	Duration	Rating	Cost

1NF: PK/ MovieID, Atomic, it is in 1NF.

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, No transitive dependencies, it is in 3NF.

4. Rental Entity

Rental (Before Normalization)						
RentalID	Rental_Date	Return_Date	Due_date	CustomerID	StaffID	Fee

Rental (After Normalization)					
RentalID(PK)	Rental_Date	Return_Date	Due_date	CustomerID	StaffID

Rental_Fee (After Normalization)		
Rental_Fee ID(PK)	RentalID(Fk)	Fee

1NF: PK/ RentalID, Atomic, it is in 1NF.

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, fee depends on RentalID, which is now separated into Rental_Fee No transitive dependencies, it is in 3NF.

5. **Payment Entity**

Payment (Before Normalization)					
PaymentID	Amount	Payment_date	Payment_method	StaffID	RentalID

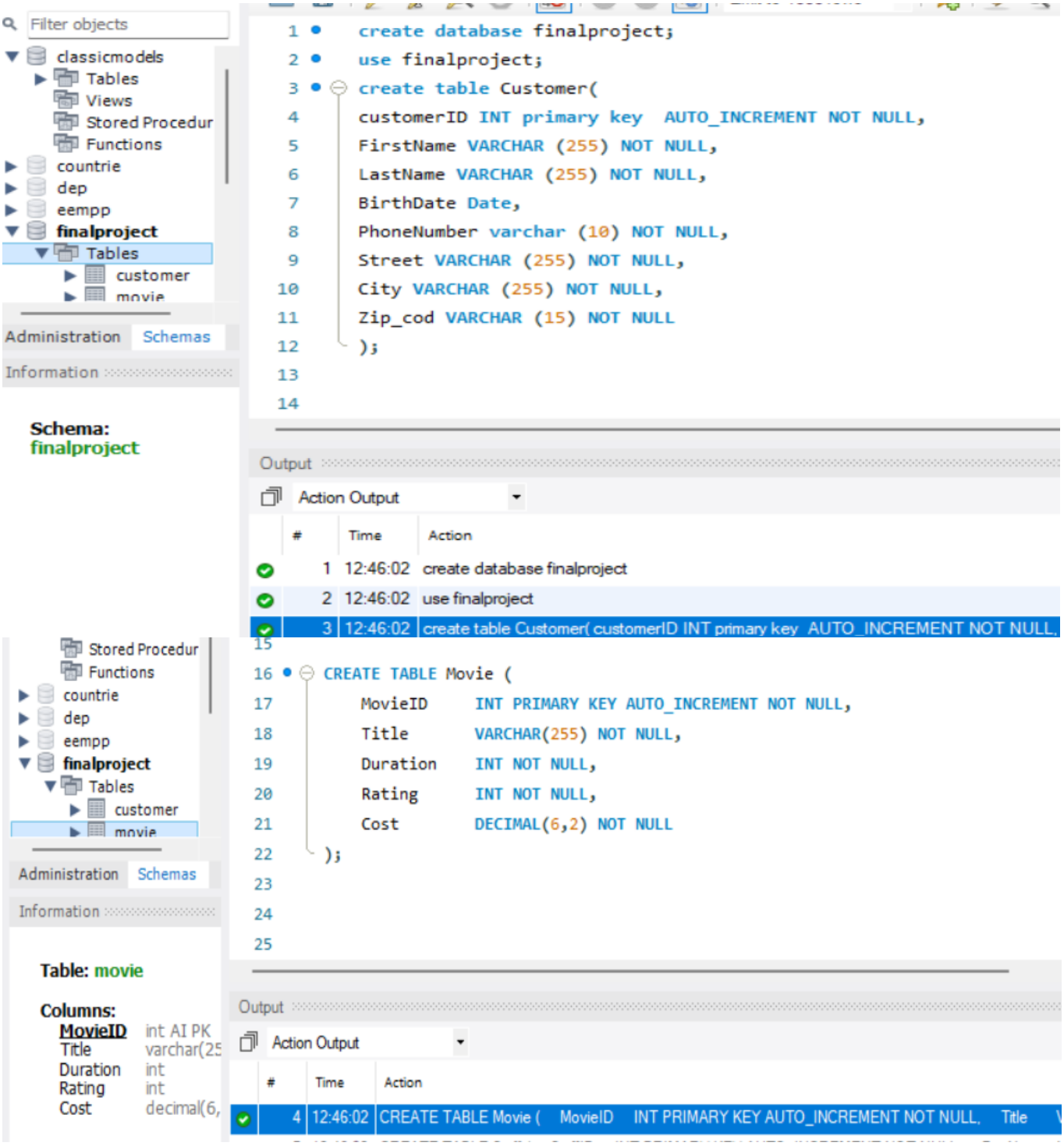
Payment (After Normalization)					
PaymentID(PK)	Amount	Payment_date	Payment_method	StaffID (FK)	RentalID(FK)

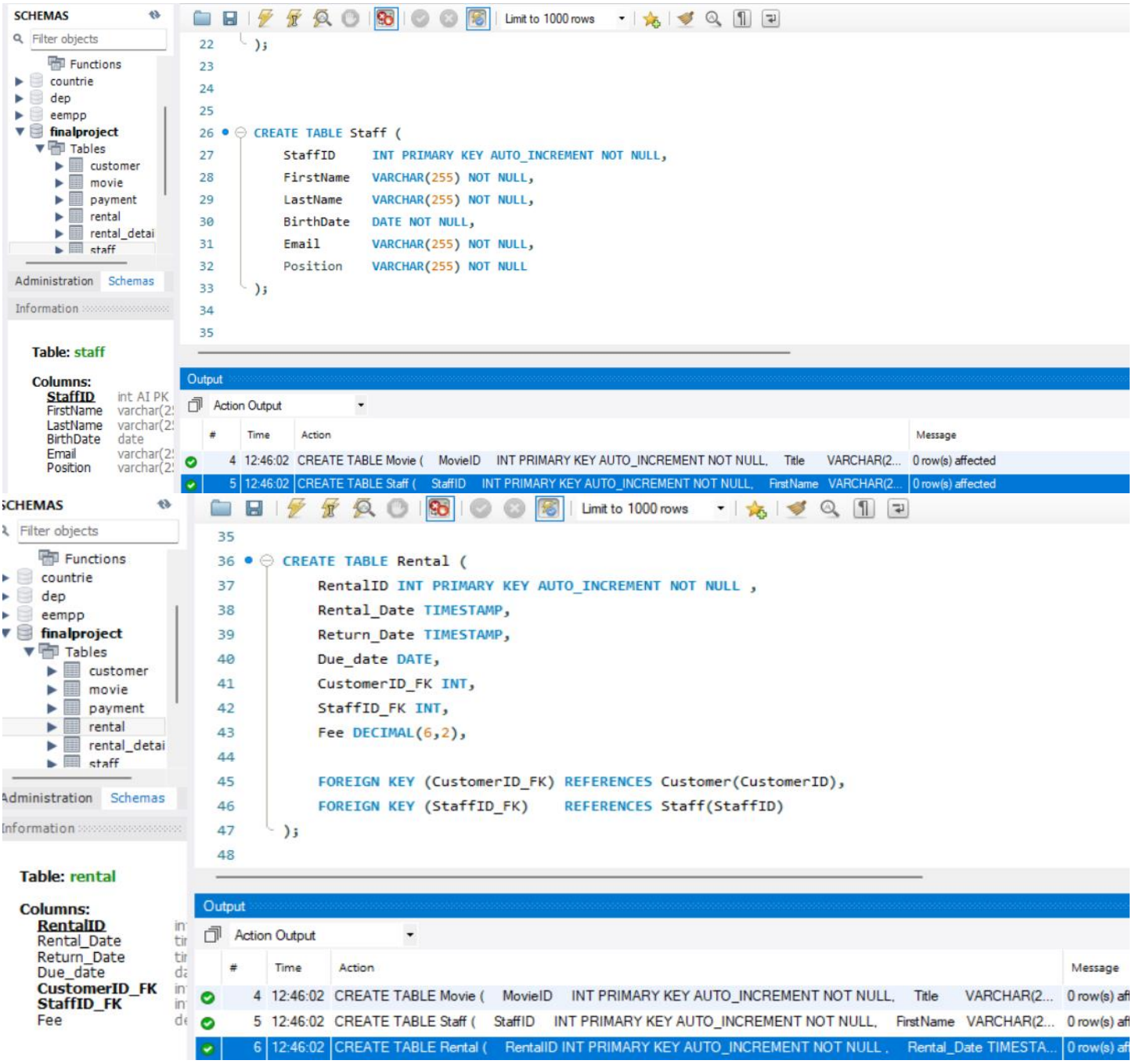
1NF: PK/ PaymentID, Atomic, it is in 1NF.

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

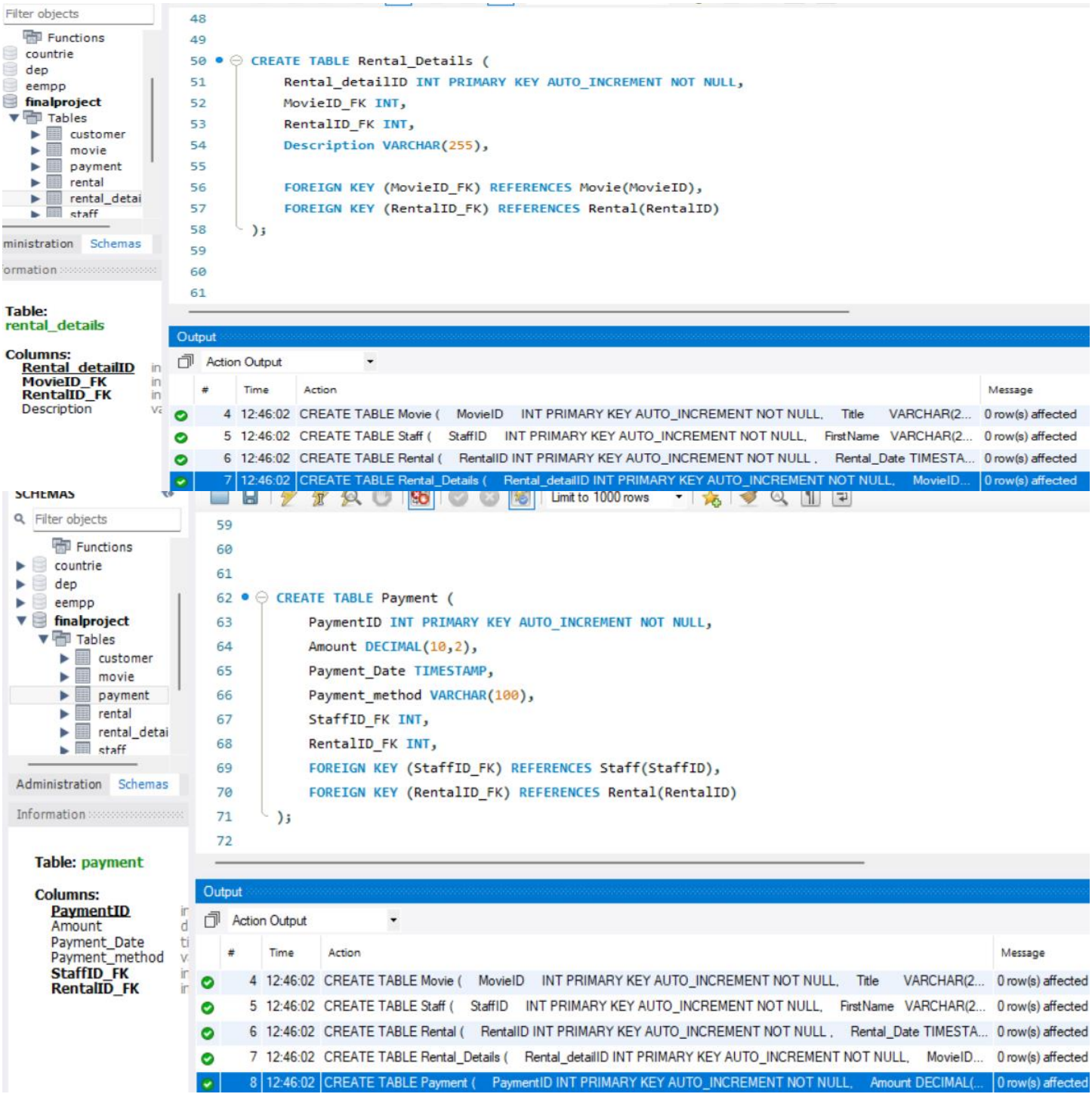
3NF: It is in 2NF, No transitive dependencies, it is in 3NF.

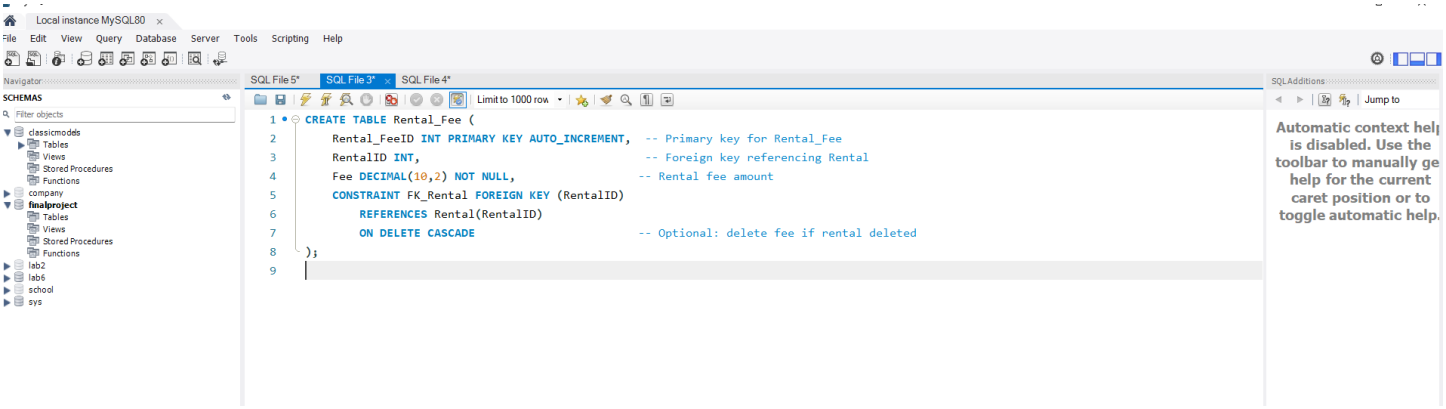
DDL implementation


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Tables
customer
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Schema:
finalproject
1 create database finalproject;
2 use finalproject;
3 create table Customer(
4 customerID INT primary key AUTO_INCREMENT NOT NULL,
5 FirstName VARCHAR (255) NOT NULL,
6 LastName VARCHAR (255) NOT NULL,
7 BirthDate Date,
8 PhoneNumber varchar (10) NOT NULL,
9 Street VARCHAR (255) NOT NULL,
10 City VARCHAR (255) NOT NULL,
11 Zip_cod VARCHAR (15) NOT NULL
12);
13
14
Output
Action Output
Time Action
1 12:46:02 create database finalproject
2 12:46:02 use finalproject
3 12:46:02 create table Customer(customerID INT primary key AUTO_INCREMENT NOT NULL,
15
16 CREATE TABLE Movie (
17 MovieID INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
18 Title VARCHAR(255) NOT NULL,
19 Duration INT NOT NULL,
20 Rating INT NOT NULL,
21 Cost DECIMAL(6,2) NOT NULL
22);
23
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Table: movie
Columns:
MovieID int AI PK
Title varchar(25
Duration int
Rating int
Cost decimal(6,
Output
Action Output
Time Action
4 12:46:02 CREATE TABLE Movie (MovieID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Title

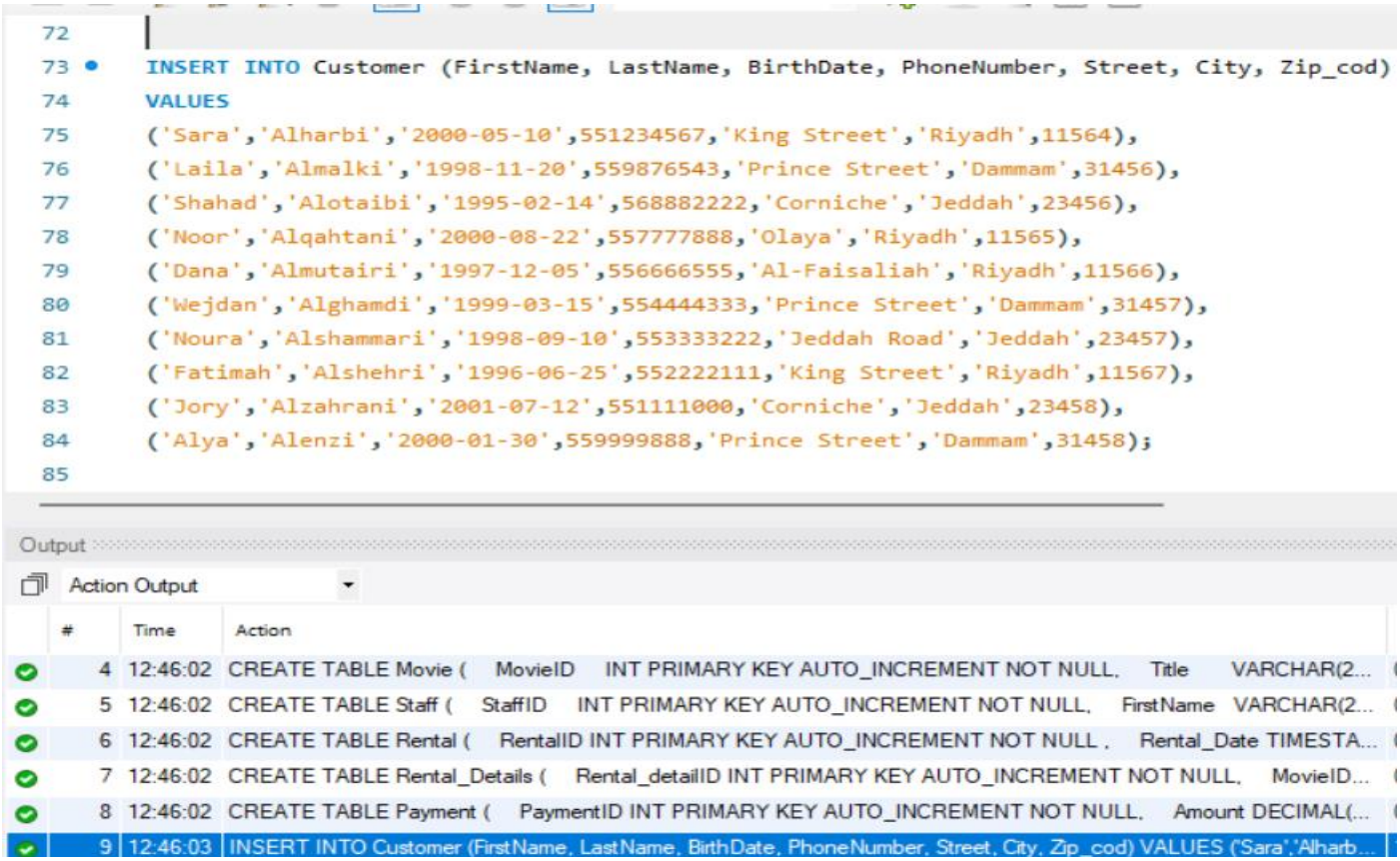

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Table: staff
Columns:
StaffID int AI PK
FirstName varchar(2
LastName varchar(2
BirthDate date
Email varchar(2
Position varchar(2
Limit to 1000 rows
22);
23
24
25
26 CREATE TABLE Staff (
27 StaffID INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
28 FirstName VARCHAR(255) NOT NULL,
29 LastName VARCHAR(255) NOT NULL,
30 BirthDate DATE NOT NULL,
31 Email VARCHAR(255) NOT NULL,
32 Position VARCHAR(255) NOT NULL
33);
34
35
Output
Action Output
Time Action Message
4 12:46:02 CREATE TABLE Movie (MovieID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Title VARCHAR(2... 0 row(s) affected
5 12:46:02 CREATE TABLE Staff (StaffID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, FirstName VARCHAR(2... 0 row(s) affected

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Information
Table: rental
Columns:
RentalID
Rental_Date
Return_Date
Due_date
CustomerID_FK
StaffID_FK
Fee
Limit to 1000 rows
35
36 CREATE TABLE Rental (
37 RentalID INT PRIMARY KEY AUTO_INCREMENT NOT NULL ,
38 Rental_Date TIMESTAMP,
39 Return_Date TIMESTAMP,
40 Due_date DATE,
41 CustomerID_FK INT,
42 StaffID_FK INT,
43 Fee DECIMAL(6,2),
44
45 FOREIGN KEY (CustomerID_FK) REFERENCES Customer(CustomerID),
46 FOREIGN KEY (StaffID_FK) REFERENCES Staff(StaffID)
47);
48
Output
Action Output
Time Action Message
4 12:46:02 CREATE TABLE Movie (MovieID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Title VARCHAR(2... 0 row(s) af
5 12:46:02 CREATE TABLE Staff (StaffID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, FirstName VARCHAR(2... 0 row(s) af
6 12:46:02 CREATE TABLE Rental (RentalID INT PRIMARY KEY AUTO_INCREMENT NOT NULL , Rental_Date TIMESTA... 0 row(s) af


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Table:
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RentalID_FK
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50 CREATE TABLE Rental_Details (
51 Rental_detailID INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
52 MovieID_FK INT,
53 RentalID_FK INT,
54 Description VARCHAR(255),
55
56 FOREIGN KEY (MovieID_FK) REFERENCES Movie(MovieID),
57 FOREIGN KEY (RentalID_FK) REFERENCES Rental(RentalID)
58);
59
60
61
Output
Action Output
Time Action Message
4 12:46:02 CREATE TABLE Movie (MovieID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Title VARCHAR(2... 0 row(s) affected
5 12:46:02 CREATE TABLE Staff (StaffID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, FirstName VARCHAR(2... 0 row(s) affected
6 12:46:02 CREATE TABLE Rental (RentalID INT PRIMARY KEY AUTO_INCREMENT NOT NULL , Rental_Date TIMESTA... 0 row(s) affected
7 12:46:02 CREATE TABLE Rental_Details (Rental_detailID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, MovieID... 0 row(s) affected
SCHEMAS
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Schemas
Information
Limit to 1000 rows
59
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61
62 CREATE TABLE Payment (
63 PaymentID INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
64 Amount DECIMAL(10,2),
65 Payment_Date TIMESTAMP,
66 Payment_method VARCHAR(100),
67 StaffID_FK INT,
68 RentalID_FK INT,
69 FOREIGN KEY (StaffID_FK) REFERENCES Staff(StaffID),
70 FOREIGN KEY (RentalID_FK) REFERENCES Rental(RentalID)
71);
72
Table: payment
Columns:
PaymentID
Amount
Payment_Date
Payment_method
StaffID_FK
RentalID_FK
Output
Action Output
Time Action Message
4 12:46:02 CREATE TABLE Movie (MovieID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Title VARCHAR(2... 0 row(s) affected
5 12:46:02 CREATE TABLE Staff (StaffID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, FirstName VARCHAR(2... 0 row(s) affected
6 12:46:02 CREATE TABLE Rental (RentalID INT PRIMARY KEY AUTO_INCREMENT NOT NULL , Rental_Date TIMESTA... 0 row(s) affected
7 12:46:02 CREATE TABLE Rental_Details (Rental_detailID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, MovieID... 0 row(s) affected
8 12:46:02 CREATE TABLE Payment (PaymentID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Amount DECIMAL(... 0 row(s) affected



DML implementation:



```sql
SELECT * FROM finalproject.customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cont

customerID	FirstName	LastName	BirthDate	PhoneNumber	Street	City	Zip_cod
1	Sara	Alharbi	2000-05-10	551234567	King Street	Riyadh	11564
2	Laila	Almalki	1998-11-20	559876543	Prince Street	Dammam	31456
3	Shahad	Alotaibi	1995-02-14	568882222	Corniche	Jeddah	23456
4	Noor	Alqahtani	2000-08-22	557777888	Olaya	Riyadh	11565
5	Dana	Almutairi	1997-12-05	556666555	Al-Faisaliah	Riyadh	11566
6	Wejdan	Alghamdi	1999-03-15	554444333	Prince Street	Dammam	31457
7	Noura	Alshammari	1998-09-10	553333222	Jeddah Road	Jeddah	23457

```sql
INSERT INTO Movie (Title, Duration, Rating, Cost)
VALUES
('The Matrix',136,5,30.00),
('Inception',148,5,35.00),
('Titanic',195,4,25.00),
('The Lion King',90,5,20.90),
('Avengers: Endgame',181,5,40.00),
('Joker',122,4,30.90),
('Frozen',102,4,25.00),
('Parasite',132,5,35.00),
('Interstellar',169,5,40.00),
('The Godfather',175,5,50.00);
```

Output

Action Output

#	Time	Action
7	12:46:02	CREATE TABLE Rental_Details (Rental_detailID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Movi...
8	12:46:02	CREATE TABLE Payment (PaymentID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Amount DECI...
9	12:46:03	INSERT INTO Customer (FirstName, LastName, BirthDate, PhoneNumber, Street, City, Zip_cod) VALUES ('Sara','Al...
10	12:46:03	INSERT INTO Movie (Title, Duration, Rating, Cost) VALUES ('The Matrix',136,5,30.00), ('Inception',148,5,35.00), ('...

```sql
SELECT * FROM finalproject.movie;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

MovieID	Title	Duration	Rating	Cost
1	The Matrix	136	5	30.00
2	Inception	148	5	35.00
3	Titanic	195	4	25.00
4	The Lion King	90	5	20.90
5	Avengers: Endgame	181	5	40.00
6	Joker	122	4	30.90
7	Frozen	102	4	25.00

movie 1

```sql
INSERT INTO Staff (FirstName, LastName, BirthDate, Email, Position)
VALUES
('Ahmed','Alqahtani','1985-05-12','ahmed.staff@example.com','Manager'),
('Sara','Alharbi','1990-08-20','sara.staff@example.com','Cashier'),
('Omar','Almutairi','1988-03-15','omar.staff@example.com','Clerk'),
('Laila','Alshehri','1992-11-10','laila.staff@example.com','Cashier'),
('Nasser','Alghamdi','1987-07-25','nasser.staff@example.com','Clerk'),
('Mona','Alzahrani','1991-09-05','mona.staff@example.com','Manager');
```

ction Output

	Time	Action
10	12:46:03	INSERT INTO Movie (Title, Duration, Rating, Cost) VALUES ('The Matrix',136,5,30.00), ('Inception',148,5,35.00), ('...
11	12:46:03	INSERT INTO Staff (FirstName, LastName, BirthDate, Email, Position) VALUES ('Ahmed','Alqahtani','1985-05-12','a...
12	12:46:03	INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK, Fee) VALUES ('2025-1

```sql
SELECT * FROM finalproject.staff;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

StaffID	FirstName	LastName	BirthDate	Email	Position
1	Ahmed	Alqahtani	1985-05-12	ahmed.staff@example.com	Manager
2	Sara	Alharbi	1990-08-20	sara.staff@example.com	Cashier
3	Omar	Almutairi	1988-03-15	omar.staff@example.com	Clerk
4	Laila	Alshehri	1992-11-10	laila.staff@example.com	Cashier
5	Nasser	Alghamdi	1987-07-25	nasser.staff@example.com	Clerk
6	Mona	Alzahrani	1991-09-05	mona.staff@example.com	Manager

```sql
SELECT * FROM finalproject.rental;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

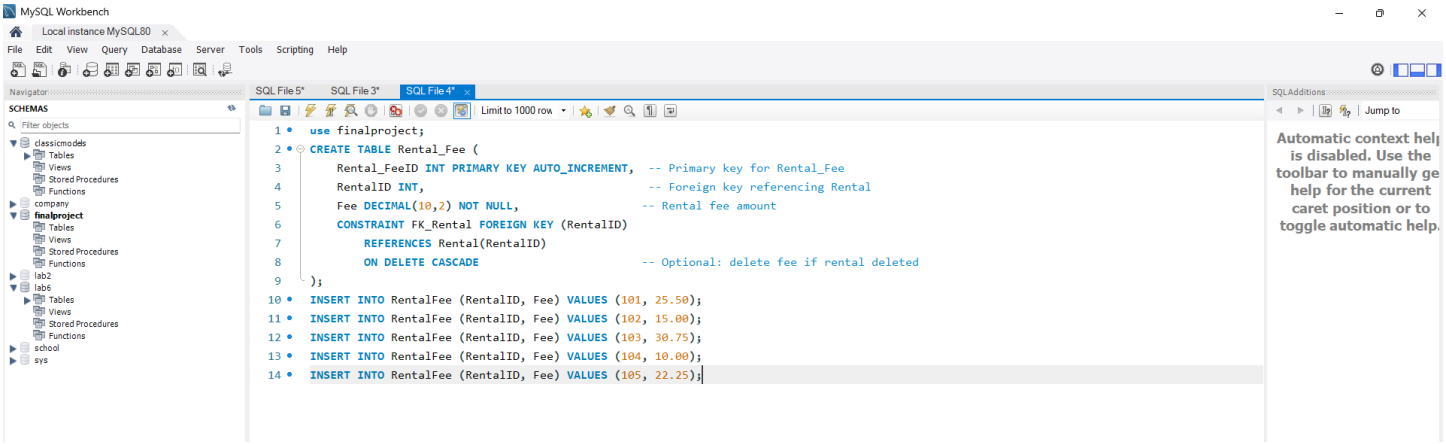
RentalID	Rental_Date	Return_Date	Due_date	CustomerID_FK	StaffID_FK	Fee
1	2025-11-01 00:00:00	2025-11-05 00:00:00	2025-11-04	1	3	5.00
2	2025-11-02 00:00:00	2025-11-08 00:00:00	2025-11-07	2	5	5.00
3	2025-11-03 00:00:00	2025-11-07 00:00:00	2025-11-07	3	2	0.00
4	2025-11-04 00:00:00	2025-11-10 00:00:00	2025-11-09	4	1	5.00
5	2025-11-05 00:00:00	2025-11-08 00:00:00	2025-11-08	5	4	0.00
6	2025-11-06 00:00:00	2025-11-12 00:00:00	2025-11-12	6	6	0.00
7	2025-11-07 00:00:00	2025-11-17 00:00:00	2025-11-16	7	3	5.00

```sql
INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK, Fee)
VALUES
('2025-11-01','2025-11-05','2025-11-04',1,3,5.00),
('2025-11-02','2025-11-08','2025-11-07',2,5,5.00),
('2025-11-03','2025-11-07','2025-11-07',3,2,0.00),
('2025-11-04','2025-11-10','2025-11-09',4,1,5.00),
('2025-11-05','2025-11-08','2025-11-08',5,4,0.00),
('2025-11-06','2025-11-12','2025-11-12',6,6,0.00),
('2025-11-07','2025-11-17','2025-11-16',7,3,5.00),
('2025-11-08','2025-11-12','2025-11-12',8,2,0.00),
('2025-11-09','2025-11-20','2025-11-19',9,5,5.00),
('2025-11-10','2025-11-20','2025-11-19',10,1,5.00);
```

```sql
INSERT INTO Payment (Amount, Payment_Date, Payment_method, StaffID_FK, RentalID_FK)
VALUES
(44.9 ,'2025-11-05','Cash',3,1),
(32.7 ,'2025-11-08','Card',5,2),
(58.4 ,'2025-11-07','Cash',2,3),
(44.9 ,'2025-11-10','Card',1,4),
(29.5 ,'2025-11-08','Cash',4,5),
(47.2 ,'2025-11-12','Card',6,6),
(61.9 ,'2025-11-17','Cash',3,7),
(33.8 ,'2025-11-12','Card',2,8),
(44.9 ,'2025-11-20','Cash',5,9),
(50.3 ,'2025-11-20','Card',1,10);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

PaymentID	Amount	Payment_Date	Payment_method	StaffID_FK	RentalID_FK
1	44.90	2025-11-05 00:00:00	Cash	3	1
2	32.70	2025-11-08 00:00:00	Card	5	2
3	58.40	2025-11-07 00:00:00	Cash	2	3
4	44.90	2025-11-10 00:00:00	Card	1	4
5	29.50	2025-11-08 00:00:00	Cash	4	5
6	47.20	2025-11-12 00:00:00	Card	6	6
7	61.90	2025-11-17 00:00:00	Cash	3	7



Triggers:

```sql
DELIMITER $$
-- Trigger to check customer>18 age
CREATE TRIGGER CheckCustomerAge
BEFORE INSERT ON Customer
FOR EACH ROW
BEGIN
    -- Calcult DateOfBirth
    IF TIMESTAMPDIFF(YEAR, NEW.BirthDate, CURDATE()) < 18 THEN
        -- Raise an error if the Customer is younger than 18
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Customer must be at least 18 years old to be added to the system.';
    END IF;
END$$
DELIMITER ;
```

```sql
DELIMITER $$
CREATE TRIGGER trgmax_rentals BEFORE INSERT ON Rental
FOR EACH ROW
BEGIN
    IF (SELECT COUNT(*)
        FROM Rental
        WHERE CustomerID_FK = NEW.CustomerID_FK and Return_Date IS NULL) >= 5 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Customer cannot have more than 5 active rentals';
    END IF;
END$$
//
DELIMITER ;
```

```sql
DELIMITER $$
CREATE TRIGGER trg_late_fee
BEFORE INSERT ON Rental
FOR EACH ROW
BEGIN
    IF NEW.Return_Date > NEW.Due_date THEN
        SET NEW.Fee = DATEDIFF(NEW.Return_Date, NEW.Due_date) * 5;
    ELSE
        SET NEW.Fee = 0;
    END IF;
END$$
DELIMITER ;
```

```sql
DELIMITER $$
CREATE TRIGGER trgmax_movie_days
BEFORE INSERT ON Rental
FOR EACH ROW
BEGIN
    IF DATEDIFF(NEW.Due_date, NEW.Rental_Date) > 10 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Movie cannot be rented for more than 10 days';
    END IF;
END$$
DELIMITER ;
```

```sql
use finalproject;
INSERT INTO Customer (FirstName, LastName, BirthDate, PhoneNumber, Street, City, Zip_cod)
VALUES ('Mona', 'Ali', '2010-05-10', '0555555555', 'King St', 'Riyadh', '11564');
```

Output

Action Output

#	Time	Action	Message
1	22:24:08	use finalproject	0 row(s) affected
2	22:24:08	INSERT INTO Customer (FirstName, LastName, BirthDate, PhoneNumber, Street, City, Zip_cod) VALUES ('Mona', 'Ali',...	Error Code: 1644. Customer must be at least 18 years old to be added to the system.

```sql
use finalproject;
INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK)
VALUES ('2025-12-01', NULL, '2025-12-05', 1, 1);
```

Output

Action Output

#	Time	Action	Message
1	22:26:49	use finalproject	0 row(s) affected
2	22:26:49	INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK) VALUES ('2025-12-01', N...	Error Code: 1644. Customer cannot have more than 5 active rentals

```sql
use finalproject;
INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK)
VALUES ('2025-12-01', NULL, '2025-12-20', 2, 1);
```

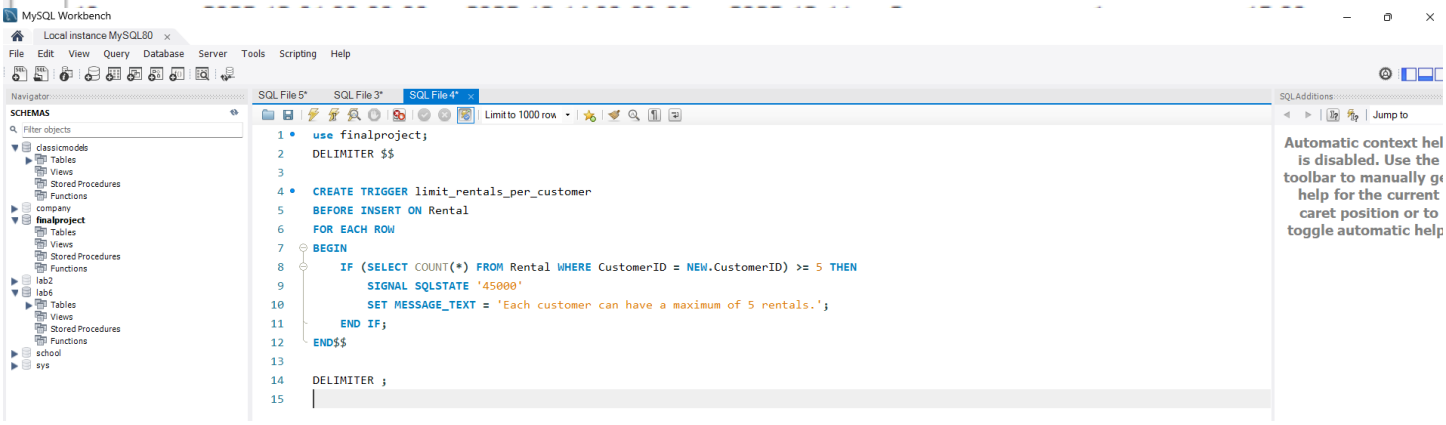
Output

Action Output

#	Time	Action	Message
1	22:27:56	use finalproject	0 row(s) affected
2	22:27:56	INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK) VALUES ('2025-12-01', N...	Error Code: 1644. Movie cannot be rented for more than 10 days

```
use finalproject;
INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK)
VALUES ('2025-12-01', '2025-12-14', '2025-12-11', 3, 1);
```

RentalID	Rental_Date	Return_Date	Due_date	CustomerID_FK	StaffID_FK	Fee
18	2025-12-01 00:00:00	2025-12-14 00:00:00	2025-12-11	3	1	15.00



Statement:

```sql
use finalproject;
SELECT FirstName, LastName, City, BirthDate
FROM Customer
WHERE (City = 'Riyadh' OR City = 'Dammam')
  AND BirthDate < '2000-01-01';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	City	BirthDate
Laila	Almalki	Dammam	1998-11-20
Dana	Almutairi	Riyadh	1997-12-05
Wejdan	Alghamdi	Dammam	1999-03-15
Fatimah	Alshehri	Riyadh	1996-06-25

```sql
use finalproject;
SELECT c.FirstName, c.LastName, m.Title, rd.Description
FROM Rental r
JOIN Customer c ON r.CustomerID_FK = c.CustomerID
JOIN Rental_Details rd ON r.RentalID = rd.RentalID_FK
JOIN Movie m ON rd.MovieID_FK = m.MovieID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	Title	Description
Sara	Alharbi	The Matrix	Rented Matrix for 4 days
Sara	Alharbi	Inception	Rented Inception for 4 days
Laila	Almalki	Titanic	Rented Titanic for 6 days
Shahad	Alotaibi	The Lion King	Rented Lion King for 4 days
Noor	Alqahtani	Avengers: Endgame	Rented Avengers for 6 days
Dana	Almutairi	Joker	Rented Joker for 3 days
Wejdan	Alghamdi	Frozen	Rented Frozen for 6 days

Aggregate Function

```sql
use finalproject;
SELECT c.FirstName, c.LastName, SUM(p.Amount) AS TotalPayments
FROM Payment p
JOIN Rental r ON p.RentalID_FK = r.RentalID
JOIN Customer c ON r.CustomerID_FK = c.CustomerID
GROUP BY c.FirstName, c.LastName;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	TotalPayments
Laila	Almalki	32.70
Shahad	Alotaibi	58.40
Noor	Alqahtani	44.90
Dana	Almutairi	29.50
Wejdan	Alghamdi	47.20
Noura	Alshammari	61.90
Fatimah	Alshehri	33.80
Jory	Alzahrani	44.90

```sql
use finalproject;
SELECT CustomerID, FirstName, LastName, BirthDate,
       TIMESTAMPDIFF(YEAR, BirthDate, CURDATE()) AS Age
FROM Customer
WHERE TIMESTAMPDIFF(YEAR, BirthDate, CURDATE()) >= 18;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerID	FirstName	LastName	BirthDate	Age
1	Sara	Alharbi	2000-05-10	25
2	Laila	Almalki	1998-11-20	27
3	Shahad	Alotaibi	1995-02-14	30
4	Noor	Alqahtani	2000-08-22	25
5	Dana	Almutairi	1997-12-05	28
6	Wejdan	Alghamdi	1999-03-15	26

```sql
SELECT CustomerID_FK AS CustomerID,
       COUNT(*) AS ActiveRentals
FROM Rental
WHERE Return_Date IS NULL
GROUP BY CustomerID_FK
HAVING COUNT(*) >= 5;
```

sult Grid | Filter Rows: | Export: | W

CustomerID	ActiveRentals
1	5

```sql
use finalproject;
SELECT RentalID, CustomerID_FK, Due_date, Return_Date, Fee
FROM Rental
WHERE Rental_Date >= '2025-12-01';
```

esult Grid | Filter Rows: | Edit: | Export/Import:

RentalID	CustomerID_FK	Due_date	Return_Date	Fee
16	3	2025-12-05	2025-12-07 00:00:00	10.00
17	4	2025-12-04	2025-12-08 00:00:00	20.00
NULL	NULL	NULL	NULL	NULL